KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2023
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours | FULL MARKS: 210

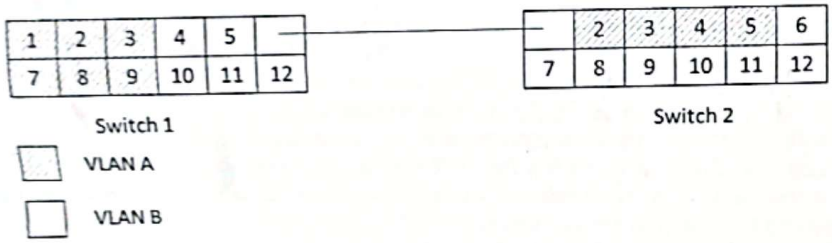
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A
(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain the purpose of OSI model. Do all devices present in a network implement all OSI layers? Justify your answer. (10) [CO1]

   b) In the network shown in following figure, the server 223.1.2.5 receives an IP packet with src: 0.0.0.0, 68 and dest: 255.255.255.255, 67. What would happen next? Complete the interaction. (12) [CO2]

   c) Explain the benefits of Port-based VLANs. Explain the process and switch table to transmit a frame from port 2 of switch 1 to port 4 of switch 2, shown in following figure. (13) [CO3]

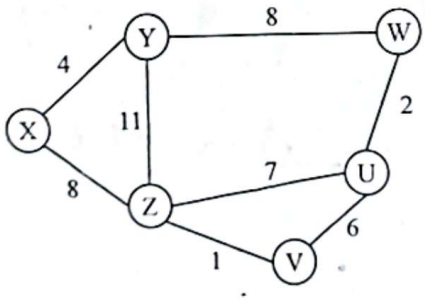


2. a) Consider the following 20-byte header (IPv4). Find the header checksum. Why does IPv4 header checksum cover only data portion? (10) [CO1]

4	5	0	28
1		0	0
4	17	0	
10.12.14.5			
12.6.7.9			

   b) Explain the hot potato routing protocol. Briefly explain the benefits and drawbacks of this protocol. (12) [CO3]

   c) Using a Link-State routing algorithm, determine the routing table at the router X from the topology given in following figure. (13) [CO2]



3. a) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to 3 group of customers: (13) [CO2]
   i) 1st group: 52 customers, each needs 128 addresses.
   ii) 2nd group: 40 customers, each needs 256 addresses.
   iii) 3rd group: 100 customers, each needs 64 addresses.

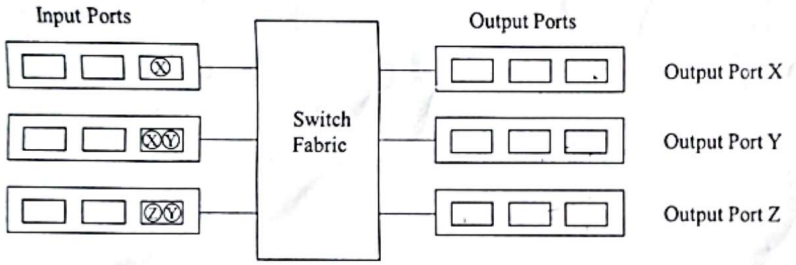
   How many addresses are still unallocated? Note down the network addressed of each group.

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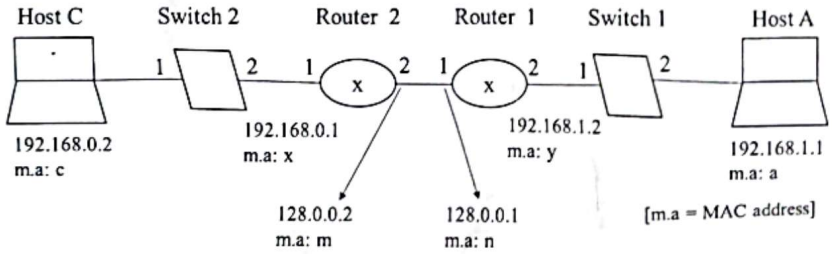
b) Suppose you have a broadcast channel in Ethernet CSMA/CD algorithm with transmission rate 10 Mbps where a node can transmit a frame of max 1500 bytes, and maximum propagation delay between 2 nodes is: 2ms. (10) [CO]
   i) Calculate the efficiency of the algorithm.
   ii) When would it reach its highest efficiency, and why?

c) In the switch shown in following figure, all datagrams have the same fixed length and the switch operates in a slotted, synchronous manner. Assuming a datagram can be transferred from input port to output port in one time slot, what is the minimal number of slots needed for the transfer of all datagrams when: (12) [CO1]
   i) The switching fabric is memory.
   ii) The switching fabric is bus.
   iii) The switching fabric is crossbar.

   Briefly explain the process for each of these switching fabrics (You can assume any input queue scheduling order).



4. a) Define SDN control plane. Briefly explain the characteristics of SDN. (10) [CO1]
   b) State router oscillation problem with proper example. When it occurs? (10) [CO3]
   c) In the network shown in following figure, a data packet is routing from A to C and C sends a response to A. Assuming all hosts know the necessary IP addresses, MAC addresses and first hop router addresses, show the data capsulation and decapsulation in each device with necessary diagram(s) and table(s). (15) [CO2]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) List down the steps involved in SSL handshaking. Also, present a pictorial view of packet transmission. (10) [CO2]
   b) Is SMTP enough to build a complete Email architecture? Justify your answer. (10) [CO2]
   c) Suppose a sender and receiver are using a pipelined protocol, i.e., Go Back-N or Selective Repeat. In this specific case, the rate at which the sender process pushes data is less than the rate at which then ~~sender~~ receiver consumes data. We want to introduce a buffer in the receiving side. Is it a wise decision? Explain the reasoning behind your answer. (05) [CO3]
   d) Let, host at *abc.poly.edu* wants IP address for *defg.cs.uxyz.edu*. Explain the procedure of getting IP address using recursive queries for various DNS servers. (10) [CO2]

6. a) State the advantage of P2P over client-server architecture. Despite those disadvantages, why do we find Client-server architecture applied in most of the real-life cases? (10) [CO1]
   b) Consider the following figure where Mr. X wants to access a webpage hosted on a Google server. This webpage comprises a base HTML file along with 10 JPEG images, each with a length of 1024 bits. The transmission rate of the link connecting Mr. X to the server is 1 Mbps. Determine the total response time for both Non-persistent and persistent HTTP by real internet delay. Mr. X measures real internet delay by traceroute program. (15) [CO2]

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```
C:\Users\HP>tracert google.com
Tracing route to google.com [142.250.193.78]
over a maximum of 30 hops:
  1    <1 ms    <1 ms    <1 ms  10.24.241.1
                  .
                  .
                  .
 14    82 ms    80 ms    80 ms  142.251.54.83
 15    80 ms    79 ms    78 ms  del11s16-in-f14.1e100.net
Trace complete
```

c) Consider a reliable data transfer protocol that uses only negative acknowledgements. Suppose the sender sends data only infrequently. Would a NAK-only protocol be preferable to a protocol that uses ACK? Why? Now suppose the sender has a lot of data to send and the end-to-end connection experiences few losses. In this second case, would a NAK only protocol be preferable to a protocol that uses ACKs? Why? Justify your answer. (10) [CO3]

7. a) Suppose, a HTTP client is sending 1 bytes of data every 5 minutes. Let, 1 HTTP packet contains 1 byte of data. The HTTP server uses the data to update its database in real time. Help the network administrator to choose between persistent and non-persistent HTTP connections. Also, advise him to use or avoid pipelining. Explain your reasoning. (15) [CO3]

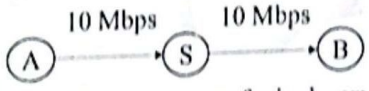
b) Suppose, in a TCP connection, the receiver process is slow at data consumption. Eventually, the receiver buffer is full and the receiver advertises zero window size. The sender halts sending. Whenever the receiver process consumes a packet, receiver sends an acknowledgement to the sender. Sender sends the next packet. Will this result in a problem? If yes, discuss about the reasoning and a possible solution. (11) [CO3]

c) State how packet filtering firewall works. (09) [CO1]

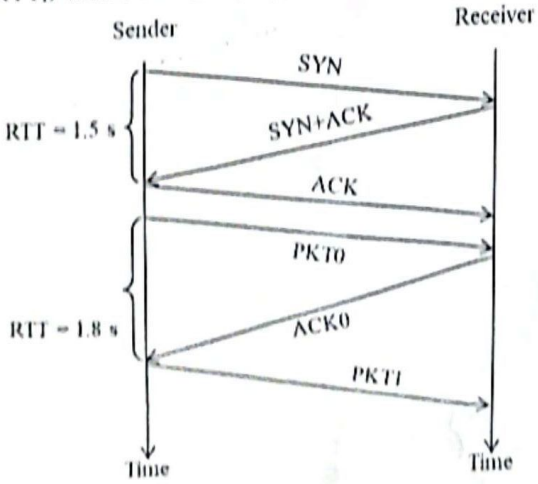
8. a) Let, a proxy server is used for web caching in case of accessing the URL *http://kuet.ac.bd/cse/home.html* and the browser of client is configured accordingly. Now, explain the scenario of accessing the web page. (10) [CO2]

b) Host A and B are each connected to a switch S via 10 Mbps link as shown in following figure. (10) [CO1]

The propagation delay on each link is 20μs. Assume signaling message for circuit setup and acceptance are 100 bit long and processing time of a signaling message at switch S, hosts A and B is 15μs. These signaling messages are transmitted over the 10 Mbps link. Calculate the total time required to transmit a 10,000-bit message from A to B for circuit switching.



c) In the following figure, some desired packet transfer is shown. Calculate Retransmission Timeout time (RTO) where applicable. Let, initial RTO=6s. Recall that, $RTO = RTT_s + 4\,RTT_D$ where $RTT_D = (1-\beta)\,RTT_D + \beta\,|RTT_s - RTT_M|$ (15) [CO3]



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